Topic

Firmware

Selections

Select Firmware Option: Node Firmware
Node Firmware Release Notes: Node Firmware 13.2.3 (Any node running OneFS 9.0.0.0 and later)

If you find any errors in this procedure or have comments regarding this application, please submit feedback here - https://contentfeedback.dell.com/home

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Contents

Preliminary Activity Tasks 3
Read, understand, and perform these tasks 3
Node Firmware 13.2.2 5

Preliminary Activity Tasks

This section may contain tasks that you must complete before performing this procedure.

Read, understand, and perform these tasks

1. Table 1 lists tasks, cautions, warnings, notes, and/or knowledgebase (KB) solutions that you need to be aware of before performing this activity. Read, understand, and when necessary perform any tasks contained in this table and any tasks contained in any associated knowledgebase solution.

Table 1 List of cautions, warnings, notes, and/or KB solutions related to this activity

Node Firmware 13.2.2

PowerScale Node Firmware Package 13.2.3

Release notes

March 2026

The firmware package is available for download from Dell Product Support. Follow the instructions in this guide to update the firmware.

NOTE: If you require assistance with the node firmware update, contact Dell Product Support and open an SR ticket.

Supported platforms and system requirements 2
Confirming free space 2
New features and resolved issues 2
Known issues 3
Firmware versions 4
Upgrade considerations 11
Where to get help 12
Additional options for getting help 12
Notes, cautions, and warnings 13

Supported platforms and system requirements

Before updating the Node Firmware Package (NFP) on the node, confirm that the OneFS version and node hardware model support the new firmware.

You must meet the following requirements for this NFP version:

- OneFS 9.0.0 and later
- Isilon Generation 6 models: A200, A2000, F800, F810, H400, H500, H600, and H5600
- PowerScale Flash models: F200, F210, F600, F710, F900, F910
- PowerScale Accelerator models: B100, P100, and PA110
- PowerScale Archive and Hybrid models: A300, A300-L, A310, A3000, A3000-L, A3100, H700, H710, H7000, and H7100

NOTE: OneFS 9.11.0.0 requires NFP 13.0 minimum.

NOTE: OneFS 9.13.0.0 requires NFP 13.2 minimum.

NOTE: For NFS over RDMA users the following versions are required: OneFS 9.2.1.15 or later, 9.3.0.8 or later, 9.4.0.5 or later, or 9.5.0.0 or later.

Confirming free space

Before upgrading the node firmware package, ensure that there is enough free space available for the node firmware binaries. Approximately 876MB of free space in /ifs/ is required, and up to 300 MB of free space in /tmp on each PowerScale Flash and Accelerator model.

New features and resolved issues

Prior NFP release notes identifying new features and resolved issues are available at the PowerScale Info Hub: https://www.dell.com/support/kbdoc/en-us/000198854/firmware-powerscale-info-hub

PowerScale flash platforms leverage the general-purpose PowerEdge server platform, customized for the PowerScale storage appliance application. Customers are advised to install only the PowerEdge firmware included in the PowerScale node firmware packages. These firmware versions are qualified within the PowerScale storage appliances, and in some instances interface dependencies require specific versions to ensure proper functionality.

New features

PowerScale Flash: F210, F710, F910, PA110

- No change from NFP 13.2.2

PowerScale Flash: F200, F600, F900, B100, P100

- No change from NFP 13.2.2

PowerScale Archive and Hybrid: A300, A310, A3000, A3100, H700, H710, H7000, H7100

- Add support for Broadcom 25GbE NICs (with compatible OneFS release)

Generation 6: A200, A2000, H400, H500, H5600, H600, F800, F810

- No change from NFP 13.2.2

Resolved issues

PowerScale Flash and Accelerator: F210, F710, F910, PA110

- None this release

PowerScale Flash and Accelerator: F200, F600, F900, B100, P100

- None this release

PowerScale Archive and Hybrid: A310, A3100, H710, H7100

- Reboot for Nvidia NIC firmware updates fails to update firmware (344545)
- Resolve BIOS to correctly report as A-rev (10736)
- Missing port information of FQDD for CX6 VPI 100GbE NIC (321280)
- Pressing x and CTRL+x have the same behavior when entering POST (10502)
- BIOS incorrectly assumes NVMe IO Queues (337891)
- Note: It is expected to encounter UEFI0074 "The Secure Boot policy has been modified since the last time the system was started" during the initial BIOS POST when upgrading the BIOS from/to version 1.0.3-1.1.0.

PowerScale Archive and Hybrid: A300, A3000, H700, H7000

- Update to Intel 2025.3 IPU
- Enable HTTP boot in BIOS (25873)
- Suppress NTB link bounce (25832)
- Add Microsoft 2023 UEFI Secure Boot certificates (26153)

Generation 6: A200, A2000, H400, H500, H5600, H600, F800, F810

- None this release

Known issues

- PowerScale flash and accelerator systems: Since the CMOS battery sensor is no longer available, the Lifecycle Controller logs display BAT0022 when the CMOS battery voltage drops below 2.2V while the server is on AC power. This log message is not available in the OneFS logging.
- The NFP install failed to make progress on the firmware update (CPE05E-280: B100, F200, F600, F700, F900, and P100)
  - This fix is pending for OneFS 9.14.0 and LTS MRs
- For PowerScale systems with CX8 400GbE dual port NIC, fans may intermittently oscillate at higher speeds (around 50% PWM) and fluctuate between different speeds even when the system is idle.

Firmware versions

The following tables list the firmware devices and versions included in the node firmware package. These tables are organized by platform type and generation.

PowerScale Accelerator and Flash Nodes

PowerScale's Accelerator and Flash nodes leverage Dell's PowerEdge server platforms and include firmware device versions qualified specifically for the PowerScale storage appliance configuration. This section identifies the firmware components managed by the Non-Disruptive Upgrade (NDU) process.

NOTE: The PowerScale 16G BIOS, CPLD, iDRAC, and VOSS firmware is qualified as a matched set, and must not be upgraded outside of the NDU process.

Table 1. PowerScale Node Firmware: F210, F710, F910, and PA110

Firmware Device	Type	Update Version	Updated in this Release
BIOS	BIOS	2.8.2	
Backplane 1 (non-F910)	FRMW	7.10	
Backplane 1 and Backplane 2 (F910 only)	FRMW	3.33	
BOSS-N1 Monolithic (PA110 only)	FRMW	2.1.13.2025	
Broadcom NetXtreme Gigabit Ethernet (BCM5720)	FRMW	22.31.6	
Integrated Remote Access Controller (iDRAC)	FRMW	7.20.80.50	
System CPLD	FRMW	1.2.7	
Emulex LPe35002-M2-D 2-Port 32Gb Fibre Channel Adapter (PA110 only)	FRMW	03.05.23	
Mellanox-CX6-200GbE	200GigE	20.39.1002+DEL0000000052	
Mellanox-CX6-100GbE	100GigE	22.32.2004+DEL0000000027	
Mellanox-CX6-25GbE	25GigE	26.36.1010+DEL0000000031	
BOSS Dell EC NVMe ISE 7450 RI M.2 80 960GB (PA110 only)	FRMW	1.1.0	
BOSS Dell NVMe ISE PE9010 GEN4 RI M.2 960GB (PA110 only)	FRMW	1.0.0	
VOSS Dell EC NVMe OPAL 7400 RI M.2 110 960GB	FRMW	1.4.0	
VOSS Dell EC NVMe FIPS 7400 RI M.2 110 960GB	FRMW	2.5.0	
VOSS Dell EC NVMe ISE 7450 RI M.2 110 960GB	FRMW	1.4.0	
VOSS Dell EC NVMe FIPS 7450 RI M.2 80 960GB	FRMW	3.4.0	
VOSS Dell EC NVMe ISE 7450 RI M.2 80 960GB	FRMW	1.4.0	
VOSS Dell EC NVMe ISE PE9010 RI M.2 80 960GB	FRMW	1.3.1	
VOSS Dell DC NVMe PM9A3 RI M.2 110 960GB	FRMW	1.1.1	
VOSS Dell EC NVMe PM9D3A RI M.2 80 960GB	FRMW	1.0.1	

Table 2. PowerScale Node Firmware: B100, F200, F600, F900, and P100

Firmware Device	Type	Update Version	Updated in this Release
BIOS	BIOS	2.25.0	
Backplane 1	FRMW	4.35	
Backplane 2 (F900 only)	FRMW	4.35	
Backplane Expander (F600 only)	FRMW	2.46	
Broadcom NetXtreme Gigabit Ethernet (BCM5720)	FRMW	22.31.6	
Dell HBA330 Mini (N/A PowerScale F900 nodes)	FRMW	16.17.01.00	
Emulex LightPulse LPe31002 (B100 only)	16Gb FC	03.05.23	
Integrated Remote Access Controller (iDRAC)	FRMW	7.00.00.183	
Intel Gigabit Ethernet (i350-t rNDC)	FRMW	22.0.9	
Mellanox-CX6-100GbE	100GigE	22.32.2004+DEL0000000027	
Mellanox-CX6-25GbE	25GigE	26.36.1010+DEL0000000031	
Mellanox-CX5-100GbE	100GigE	16.35.1012+DEL0000000004	
Mellanox-CX4-25GbE	25GigE	14.32.2004+DEL2420110034	
Mellanox-CX3	IB/40 GbE	2.42.5000+EMC0000000004	

PowerScale Archive and Hybrid Nodes

This section identifies the firmware components managed by the Non-Disruptive Upgrade (NDU) process.

Table 3. PowerScale Archive and Hybrid Node Firmware: A310, A3100, H710, and H7100

Firmware Device	Purpose / Type	Update Version	Updated in this Release
AltoSaxadpt_symphony	BMC - Adaptive Cooling	03.70	
AltoSaxbem_as_mongoose_lw_m1b	Battery Charge Circuit	03.20	
AltoSaxgem_symphony	BMC - Main (GEM)	11.01	Yes
AltoSaxccdes_catamarancos	Catamaran CDES V2	1.10.8+21.00	
AltoSaxast26xx_cec_region0	Cryptographic Embedded Controller	03.00	
AltoSaxast26xx_cec_region1	Cryptographic Embedded Controller	32.03	
AltoSaxast26xx_cec_region3	Cryptographic Embedded Controller	00.03	
AltoSaxcmd_rev_table	CMD Revision Table (not software)	20.84	
AltoSaxcmd_100dc12pin_v1	CMD Image (motherboard)	02.12.75.03	
AltoSaxcmd_64pincos	CMD Image (Catamaran V2)	02.09.117.03	
AltoSaxeth_bcm5720_altosax	1GbE LAN FW	01.00	
AltoSaxfp_fpinffw	Front Panel LED Infinity	02.0c	
AltoSaxfp_fpioc	Front Panel LED Infinity	00.0b	
AltoSaxfw_map_table	Firmware Map Table (not software)	00.04	
AltoSaxstoragebios_altosax	BIOS	01.10	Yes
AltoSaxps0_gen3_acbel_optimuswr	CFF PSU	03.01	
AltoSaxps0_gen3_artesyn_ramjet	CFF PSU	02.30	
AltoSaxps0_gen3_liteon_atlas	CFF PSU	03.08	
AltoSaxps1_gen3_acbel_optimuswr	CFF PSU	03.01	
AltoSaxps1_gen3_artesyn_ramjet	CFF PSU	02.30	
AltoSaxps1_gen3_liteon_atlas	CFF PSU	03.08	
AltoSaxsas_catamaran_3808	SAS controller (Catamaran V2)	0028.0015.0000.0000	
AltoSaxssp_altosax	BMC - SSP	02.00	
AltoSaxvrd0_altosax_inf	Voltage Regulator	00.05	
AltoSaxvrd0_altosax_inf_v1	Voltage Regulator	02.03	
AltoSaxvrd0_altosax_rea	Voltage Regulator	00.20	
AltoSaxvrd0_altosax_rea_v1	Voltage Regulator	02.20	
AltoSaxaltosax	SP SPI ROM	01.10	Yes
M.2-NVMe-7400-FIPS	M.2 drive firmware	2.5.0	
M.2-NVMe-7450-FIPS	M.2 drive firmware	3.1.0	
M.2-NVMe-7450-ISE	M.2 drive firmware	1.1.0	
Network Interface Cards	Mellanox CX6-25GbE	26.36.1010+DEL0000000031	
	Mellanox CX6-100GbE	22.32.2004+DEL0000000027	
	Mellanox CX6 IB/100GbE	20.37.1014+EMC0000000029	

Table 4. PowerScale Archive and Hybrid Node Firmware: A300, A3000, H700, and H7000

Firmware Device	Purpose / Type	Update Version	Updated in this Release
MLKadpt_skylanders	BMC - Adaptive Cooling	02.80	
MLKbem_mongoose	Battery Charge Circuit	03.20	
MLKbem_mongoose_lw_mlb	Battery Charge Circuit	03.20	
MLKcatamaran	SAS controller (Catamaran V2)	0028.0015.0000.0000	
MLKccdes	Catamaran CDES	1.10.8+21.00	
MLKccdes_catamaran	Catamaran CDES V2	1.10.8+21.00	
MLKcdes	CDES-2 (Gen6-MLK)	2.44.0+0.18.0+21.00	
MLKcec_region0	Cryptographic Embedded Controller	02.00	
MLKcec_region1	Cryptographic Embedded Controller	32.02	
MLKcec_region3	Cryptographic Embedded Controller	00.03	
MLKcmd_rev_table	CMD Revision Table (not software)	20.85	
MLKcmd_100dc12pin	CMD Image	02.13.75.03	
MLKcmd_64pin	CMD Image	02.05.102.02	
MLKcmd_64pincos	CMD Image (Catamaran V2)	02.09.117.03	
MLKbios_gryphon	BIOS	10.81	Yes
MLKpost_gryphon	POST	10.43	
MLKserdes_infinitymlk	SERDES	00.50	
MLKspime_gryphon	BIOS ME	10.00	Yes
MLKuefi_fw_volume	UEFI	40.87	
MLKeth_infinitymlk	1GbE LAN FW	02.64	
MLKfp_fpinffw	Front Panel LED Infinity	02.0c	
MLKfp_fpioc	Front Panel LED Infinity	00.0b	
MLKfullbios_gryphon_post	SP SPI ROM	10.81	Yes
MLKfw_map_table	Firmware Map Table (not software)	40.52	
MLKgem_skylanders	BMC - Main (GEM)	45.66	
MLKps0_gen2_artesyn_optimus	CFF PSU	02.18	
MLKps0_gen2_artesyn_ramjet	CFF PSU	02.18	
MLKps0_gen3_acbel_optimuswr	CFF PSU	03.01	
MLKps0_gen3_artesyn_ramjet	CFF PSU	02.30	
MLKps1_gen2_artesyn_optimus	CFF PSU	02.18	
MLKps1_gen2_artesyn_ramjet	CFF PSU	02.18	
MLKps1_gen3_acbel_optimuswr	CFF PSU	03.01	
MLKps1_gen3_artesyn_ramjet	CFF PSU	02.30	
MLKsas_catamaran	Gen6 SAS controller UEFI driver	0015.0004.0000.0000	
MLKsas_infinitymlk	SAS controller	0001.0004.0000.11193	
MLKssp_infinitymlk	BMC - SSP	01.44	
MLKvrd0_infinitymlk_infineon	Voltage Regulator	01.01	
MLKvrd1_infinitymlk_infineon	Voltage Regulator	01.01	
MLKvrd2_infinitymlk_infineon	Voltage Regulator	01.01	
MLKvrd3_infinitymlk_infineon	Voltage Regulator	01.02	
MLKvrd0_infinitymlk_TI	Voltage Regulator	01.01	
MLKvrd1_infinitymlk_TI	Voltage Regulator	01.01	
MLKvrd2_infinitymlk_TI	Voltage Regulator	01.01	
MLKvrd3_infinitymlk_TI	Voltage Regulator	01.01	
Network Interface Cards	Mellanox CX3 IB/40GbE	2.42.5000+EMC0000000004	
	Mellanox CX4-25GbE	14.32.2004+DEL2420110034	
	Mellanox CX5-100GbE	16.35.1012+DEL0000000004	
	Mellanox CX6-25GbE	26.36.1010+DEL0000000031	
	Mellanox CX6-100GbE	22.32.2004+DEL0000000027	
	Mellanox CX6 IB/40GbE	20.37.1014+EMC0000000029	

Isilon Generation 6 Nodes

This section identifies the firmware components managed by the Non-Disruptive Upgrade (NDU) process.

Table 4. Isilon Generation 6 Node Firmware: A200, A2000, H400, H500, H5600, H600, F800, and F810

Isilon Platform	Firmware Device	Update Version	Updated in this Release
A200, A2000, and H400	DEadpt_moons	01.80	
	DEbcc_infinity	Replaced by DEbem_mongoose_lw and DEbem_mongoose	
	DEbem_mongoose_lw (bcc f/w)	03.20	
	DEbem_mongoose (bcc f/w)	03.20	
	DEbios_banshee	37.44	
	DEbmc_bblock_moons	01.40	
	DEbmc_main	24.52	
	DEccdes	1.10.8+21.00	
	DEcdes	2.44.0+0.18.0+21.00	
	DEcmd_100d815pin	03.08.103.01	
	DEcmd_mam_64pin	02.05.102.02	
	DEcmd_cat_64pin	02.05.102.02	
	DEeth_infinity	02.52	
	DEeth_sageville_infinity	0128.0000.0005.0195	
	DEfp_fpinffw	02.0c	
	DEfp_fpioc	00.0b	
	DEpost_banshee	28.18	
	DEps	gen2_artesyn: 02.14 zeus_acbel: 04.12 gen3_acbel_optimuswr: 03.01 gen3_artesyn_ramjet: 02.30	
	DEsas_catamaran	0015.0004.0000.0000	
	DEsas_infinity	0001.0004.0000.11193	
	DEserdes_banshee	08.00	
	DEspime_banshee	04.00	
	DEssp_infinity	02.84	
	DEuefi_fw_volume_infinity	03.43	
	DEvrd0_banshee_stmicro	02.01	
	DEvrd1_banshee_stmicro	01.00	
	DEps0_gen2_acbel_hv_zeus	04.12	
	DEps0_gen2_artesyn	02.18	
	DEps0_gen3_acbel_optimuswr	03.01	
	DEps0_gen3_artesyn_ramjet	02.30	
	DEps1_gen2_artesyn	02.18	
	DEps1_gen3_acbel_optimuswr	03.01	
	DEps1_gen3_artesyn_ramjet	02.30	
H500, H5600, H600, F800, and F810	EPadpt_moons	01.80	
	EPbcc_infinity	Replaced by EPbem_mongoose_lw and EPbem_mongoose	
	EPbem_mongoose_lw (bcc f/w)	03.20	
	EPbem_mongoose (bcc f/w)	03.20	
	EPbios_warthog	32.54	
	EPbmc_bblock_moons	01.40	
	EPbmc_main	24.52	
	EPccdes	1.10.8+21.00	
	EPcdes	2.44.0+0.18.0+21.00	
	EPcmd_100d815pin	03.08.103.01	
	EPcmd_mam_64pin	02.05.102.02	
	EPcmd_cat_64pin	02.05.102.02	
	EPeth_infinity	02.52	
	EPfp_fpinffw	02.0c	
	EPfp_fpioc	00.0b	
	EPpost_warthog	22.17	

Isilon Platform	Firmware Device	Update Version	Updated in this Release
	EPps	gen2_artesyn: 02.14 zeus_acbel: 04.12 gen3_acbel_optimuswr: 03.01 gen3_artesyn_ramjet: 02.30	
	EPsas_catamaran	0015.0004.0000.0000	
	EPsas_infinity	0001.0004.0000.11193	
	EPserdes_warthog	08.00	
	EPspime_warthog	10.00	
	EPssp_infinity	02.84	
	EPuefi_fw_volume_infinity	03.43	
	EPvrd0_warthog_stmicro	01.01	
	EPvrd1_warthog_stmicro	01.01	
	EPvrd2_warthog_stmicro	01.01	
	EPps0_gen2_acbel_hv_zeus	04.12	
	EPps0_gen2_artesyn	02.18	
	EPps0_gen3_acbel_optimuswr	03.01	
	EPps0_gen3_artesyn_ramjet	02.30	
	EPps1_gen2_acbel_hv_zeus	04.12	
	EPps1_gen2_artesyn	02.18	
	EPps1_gen3_acbel_optimuswr	03.01	
	EPps1_gen3_artesyn_ramjet	02.30	
Network Interface Cards	Emulex LightPulse FC Adapter (Fibre Channel + 10GbE)	11.4.204.21	
	Mellanox CX-3 IB/10GbE	2.42.5000+ISL1090110018	
	Mellanox CX-3 IB/40GbE	2.42.5000+EMC0000000004	
	Mellanox CX-3 (40GbE)	2.42.5000+EMC1090111023	
	Mellanox CX-3 (10GbE)	2.42.5000+EMC0000000002	
	Mellanox CX-4 (25GbE)	14.32.2004+DEL2420110034	
	Innova 40GbE Compression NIC (F810 only)		
	FPGA	0x100a+0x100a	
	Firmware	16.28.1002+EMC0000000017	

Upgrade considerations

Before you begin a firmware upgrade, be aware of the effect the upgrade process has on cluster operations.

Each node is upgraded and can restart automatically when the upgrade is complete. There are three upgrade options: parallel (recommended), rolling (slowest), or simultaneous (fastest). The time taken for upgrade is reduced substantially by using the simultaneous option. Simultaneous upgrades update the cluster firmware in less time than parallel or rolling upgrades. However, there is slightly more risk for client disruption during a simultaneous upgrade.

NOTE: If there is a partial upgrade, the system generates an alert indicating that a firmware upgrade has not been applied.

Run a firmware assessment

The firmware upgrade process uses the upgrade logs for firmware assessments. It summarizes data gathered by `isi_upgrade_logs` and provides that information in a report.

The Node Firmware Package must be in the /ifs directory.

1. To run an assessment, enter the following command:
   ```
   isi upgrade firmware assess --fw-pkg=/ifs/<path to NFP>
   ```
2. To see the results and obtain the report, enter the following command:
   ```
   isi_upgrade_logs --get-fw-report
   ```
3. Follow the recommendations in the report.

The assessment might take an extended amount of time to complete based on cluster size, activity, platform types, etc.

Upgrade the firmware

Upgrade nodes to the current firmware package.

To determine which nodes require upgrading, view the firmware assessment report.

1. Follow the recommendations in the firmware assessment report.
2. To upgrade the nodes to the current firmware package, run the following command:
   ```
   isi upgrade cluster firmware start --fw-pkg=/ifs/<path to nfp> --nodes-to-upgrade=<node number> <--
   rolling | --parallel | --simultaneous>
   ```

   NOTE: You must choose either the rolling, simultaneous, or parallel upgrade option for the command to function. If you are unsure which option to choose, --parallel is the recommended option.

3. To check the progress of the firmware upgrade, run the following command at any time:
   ```
   isi upgrade cluster nodes firmware progress list
   ```
4. When the upgrade completes, run the firmware assessment report and if any nodes still need an upgrade, repeat steps 1 and 2 on those nodes.
5. The firmware upgrade is complete when the firmware assessment report does not list any nodes that need upgrades.

Where to get help

The Dell Technologies Support site contains important information about products and services including drivers, installation packages, product documentation, knowledge base articles, and advisories.

A valid support contract and account might be required to access all the available information about a specific Dell Technologies product or service.

Additional options for getting help

This section contains resources for getting answers to questions about PowerScale products.

Dell Technologies support	• https://www.dell.com/support/incidents-online/contactus/product/isilon-onefs.
Telephone support	• United States: 1-800-SVC-4EMC (1-800-782-4362) • Canada: 1-800-543-4782 • Worldwide: 1-312-725-5401 • Local phone numbers for a specific country or region are available at https://www.dell.com/support/incidents-online/contactus/product/isilon-onefs.
PowerScale OneFS Documentation InfoHubs	• https://www.dell.com/support/kbdoc/000152189/powerscale-onefs-info-hubs
Dell Community Board for self-help	• https://www.dell.com/community

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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